

Architecture Decision Records (ADRs)

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System: Case Ace v2.0 Client Consultation Recording & Note Drafting System

Organisation: Citizens Advice Wandsworth (CAW)

Status: Formally Approved Baseline Architecture

Classification: Internal / Technical Reference

Index of Architectural Decisions

ADR-001: In-Browser Volatile-Only Memory Architecture (Eliminating Client-Side Disk Persistence)

ADR-002: Local-First Whisper WASM for Identifier Extraction

ADR-003: Acoustic Muting & Verified Audio Egress (Zero Raw Audio to Cloud)

ADR-004: Multi-Layer Hybrid Named Entity Recognition (Regex + Contextual NER + Special Category)

ADR-005: Surrogate Tokenisation & In-Browser Reverse Detokenisation for LLM Drafting

ADR-006: Dual-Pass Speech-to-Text Pipeline (Local Whisper + Cloud Chirp 2)

ADR-007: Single-Function Deterministic Session Destruction (destroySession())

ADR-008: Strict Whitelist-Only Telemetry Logging (Zero Free-Text / Zero PII Storage)

ADR-009: Pinning Cloud Sub-Processors Strictly to UK Sovereign Region (eu-west-2 London)

ADR-010: In-Browser SRTP Telephony Decryption for Cisco Webex Integration

ADR-001: In-Browser Volatile-Only Memory Architecture (Eliminating Client Disk Persistence)

Status: ACCEPTED (Phase 1)

Context: UK GDPR Article 5(1)(e) (Storage Limitation) and Article 25 (Data Protection by Design) require minimizing the physical footprint of sensitive advice audio. Saving audio to browser storage (localStorage, sessionStorage, IndexedDB) risks data recovery after browser crashes or computer reassignment.

Decision: All consultation audio, intermediate transcripts, and drafts are stored strictly in volatile JavaScript ArrayBuffer / Float32Array heap memory. Direct calls to non-volatile browser storage APIs are blocked at build time via automated CI AST linting (scripts/lint-storage-guard.mjs).

Consequences: Zero client consultation audio persists to disk. Session state is ephemeral; browser refresh or tab closure instantly frees memory.

ADR-002: Local-First Whisper WASM for Identifier Extraction

Status: ACCEPTED (Phase 8)

Context: Identifying names, addresses, and NINOs requires speech recognition. Sending raw audio to cloud ASR before redaction violates the Zero-Unredacted-Audio egress constraint.

Decision: Execute a quantized local Whisper ASR model (whisper-base-en-v3 / whisper.cpp) compiled to WebAssembly (WASM) with SIMD acceleration running in a dedicated Web Worker within the adviser's browser.

Consequences: Fast preliminary transcription occurs 100% locally on the device without network transmission.

ADR-003: Acoustic Muting & Verified Audio Egress

Status: ACCEPTED (Phase 10)

Context: Muting personal identifiers in the audio stream must be fail-safe before any redacted WAV file is sent to Google Cloud STT v2 for high-fidelity transcription.

Decision: Audio samples corresponding to identified entity timestamps are zeroed in-place (Uint8Array.fill(0)). A secondary local ASR verification pass transcribes the muted audio stream. If any identifier sound survives, cloud transmission is blocked immediately.

Consequences: Guarantee that zero audible personal identifiers leave the local browser boundary.

ADR-004: Multi-Layer Hybrid Named Entity Recognition (NER)

Status: ACCEPTED (Phase 8)

Context: Single NER engines suffer from blind spots across structured codes (NINOs, postcodes) vs free-text names, employers, and special category disclosures (safeguarding, medical diagnoses).

Decision: Implement a 3-layer sequential NER pipeline:

Layer 1: Deterministic Regex & Checksum Validators (NINO, UK Postcodes, Phone numbers, DOBs).

Layer 2: Contextual Gazetteer & Rule-based NER (Names, Creditors, Landlords, Employers).

Layer 3: Special Category Health & Safeguarding Classifier.

Consequences: Measured empirical recall $\geq 92.3\%$ on the 33-scenario synthetic benchmark.

ADR-005: Surrogate Tokenisation & In-Browser Reverse Detokenisation

Status: ACCEPTED (Phase 13)

Context: Sending unredacted text transcripts to Large Language Models (LLMs) exposes client PII to cloud endpoints.

Decision: Replace all confirmed personal entities with surrogate tokens ([CLIENT_NAME_1], [POSTCODE_1], [NINO_1]) prior to cloud transmission. The LLM (Gemini 1.5 in London) reasons over surrogate tokens and produces a structured AQS note. The browser substitutes real entities back into the draft note locally.

Consequences: Zero direct client PII is exposed to the cloud drafting LLM.

ADR-006: Dual-Pass Speech-to-Text Pipeline

Status: ACCEPTED (Phase 10)

Context: Local Whisper WASM provides initial redaction bounding, but high-volume casework requires 99%+ transcription accuracy across heavy regional dialects and complex benefit terminology.

Decision: Combine local Whisper WASM (Pass 1 - Entity Detection) with Google Cloud STT v2 chirp_2 / latest_long in europe-west2 (Pass 2 - Verified Redacted Audio Transcription).

Consequences: Combines local privacy boundaries with cloud acoustic accuracy.

ADR-007: Single-Function Deterministic Session Destruction (destroySession())

Status: ACCEPTED (Phase 15)

Context: Multiple exit paths (logout, timeout, tab close, consent withdrawal, error) could lead to orphaned state if destruction logic is fragmented.

Decision: Centralize all teardown logic into a single canonical, non-reentrant destroySession() function. Overwrites all TypedArray audio buffers with zeroes (.fill(0)), clears token maps, terminates recovery workers, revokes cloud credentials, and wipes the clipboard.

Consequences: Guaranteed clean state across all 6 exit paths.

ADR-008: Strict Whitelist-Only Telemetry Logging

Status: ACCEPTED (Phase 16)

Context: Telemetry logs must support operations and security audits without creating a secondary data leak vector.

Decision: Implement a strict JSON schema validator (logSchema.ts) that validates telemetry against a strict whitelist of integer counts, enums, durations, and pseudonymous IDs. Rejects any unapproved or free-text fields. Enforces 365-day automated TTL purging.

Consequences: Telemetry logs are incapable of storing client narrative or identifying details.

ADR-009: Pinning Cloud Sub-Processors Strictly to UK Sovereign Region (europe-west2)

Status: ACCEPTED (Phase 12)

Context: Post-Brexit UK GDPR International Data Transfer requirements require strict control over cross-border data flows.

Decision: All cloud endpoints (Google Cloud Speech-to-Text v2 and Vertex AI Gemini) are pinned strictly to europe-west2 (London). Multi-region failover to US or EU regions is disabled.

Consequences: Complete UK data residency; zero international data transfers.

ADR-010: In-Browser SRTP Telephony Decryption for Cisco Webex

Status: ACCEPTED (Phase 5)

Context: Capturing Webex telephone consultations must avoid cloud server-side recording.

Decision: Case Ace integrates with Cisco Webex via OAuth (spark:calls_read, spark:kms) and WebRTC. Audio RTP streams are encrypted end-to-end and decrypted locally in the adviser's browser sandbox via WebRTC SRTP keys. Cloud recording on Webex is permanently disabled.

Consequences: Telephone consultation audio is captured exclusively in local volatile browser memory.